Specification Sheet

Date	27/10/2026
Type	Exercise

- urban data sources:
  - city model: https://geodaten.bayern.de/opengeodata/OpenDataDetail.html?pn=lod2
  - city models in CITYGML format
- Project Name: Passau Flood Explorer
- Project Goal:
  - flood simulation/ visualize rising water levels in the Danuber, Inn, and Ilz and how it affects the city of Passau
  - understand flood development
  - identify potentially affected areas
  - explore and analyze historical flood events
  - Bavarian water authority's feasibility study for Passau uses the planning principle "100-year flood = 1954 flood + 0.6 m"
    → explore protection scenarios
- Datasets:
  - OpenData LoD2 city model of Passau → format: CITYGML
  - water levels of Passau → https://www.hnd.bayern.de/warnungen?id=275&warnungstyp=lkr
- Devices
  - Pimax Crystal Super for VR/AR

    - Desktop workstation
- Visualization techniques:
    - **Desktop user (analyst):**
        - overview, queries, filters, annotations, charts (scatterplot), and comparisons across districts or time
        - picks a scenario (gauge level, warning level, or a historical event) and filters buildings, for example by function, floor area, or land use, to see which are affected
        - annotate/mark specific buildings/areas
    - **VR user (explorer):**
        - 3D Surface Rendering of the city
        - embodied, spatial exploration at human scale, such as walking through a street, judging sightlines, and checking how a building feels in context
        - stands in the old town at real scale while a water level rises, and judges how deep the water would be at specific places, which routes would be blocked, and which buildings are affected
        - mark critical spots that show up on the desktop map
        - mark positions for potentially useful protection barriers for future floodings
        - interaction techniques:
            - explore by walking/flying
            - mark special positions/critical areas
    - **Collaboration aspects:**
        - annotations, filters made in 2D are also highlighted in the virtual environment using 3D color mapping
        - marks made in the virtual environment are also displayed in the 2D interface

- Desktop user sees VR user’s position and markers